

Antioxidant and antiproliferative activities of common fruits.

Consumption of fruits and vegetables has been associated with reduced risk of chronic diseases such as cardiovascular disease and cancer. Phytochemicals, especially phenolics, in fruits and vegetables are suggested to be the major bioactive compounds for the health benefits. However, the phenolic contents and their antioxidant activities in fruits and vegetables were underestimated in the literature, because bound phenolics were not included. This study was designed to investigate the profiles of total phenolics, including both soluble free and bound forms in common fruits, by applying solvent extraction, base digestion, and solid-phase extraction methods. Cranberry had the highest total phenolic content, followed by apple, red grape, strawberry, pineapple, banana, peach, lemon, orange, pear, and grapefruit. Total antioxidant activity was measured using the TOSC assay. Cranberry had the highest total antioxidant activity (177.0 $\pm$ 4.3 micromol of vitamin C equiv/g of fruit), followed by apple, red grape, strawberry, peach, lemon, pear, banana, orange, grapefruit, and pineapple. Antiproliferation activities were also studied in vitro using HepG(2) human liver-cancer cells, and cranberry showed the highest inhibitory effect with an EC(50) of 14.5 $\pm$ 0.5 mg/mL, followed by lemon, apple, strawberry, red grape, banana, grapefruit, and peach. A bioactivity index (BI) for dietary cancer prevention is proposed to provide a new alternative biomarker for future epidemiological studies in dietary cancer prevention and health promotion.